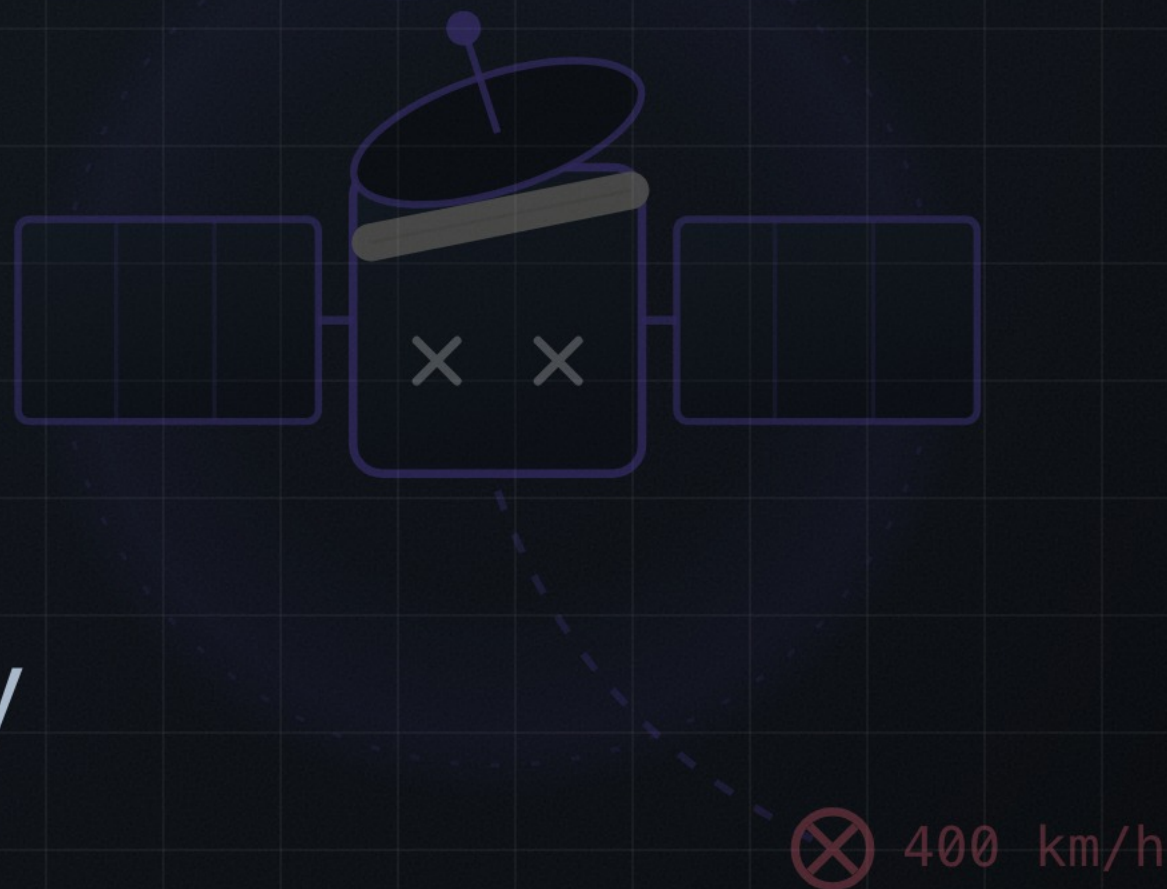


Your GPS is lying to you

400

Our app once clocked a
parked car at 400 kmph.
The phone was completely
sure.

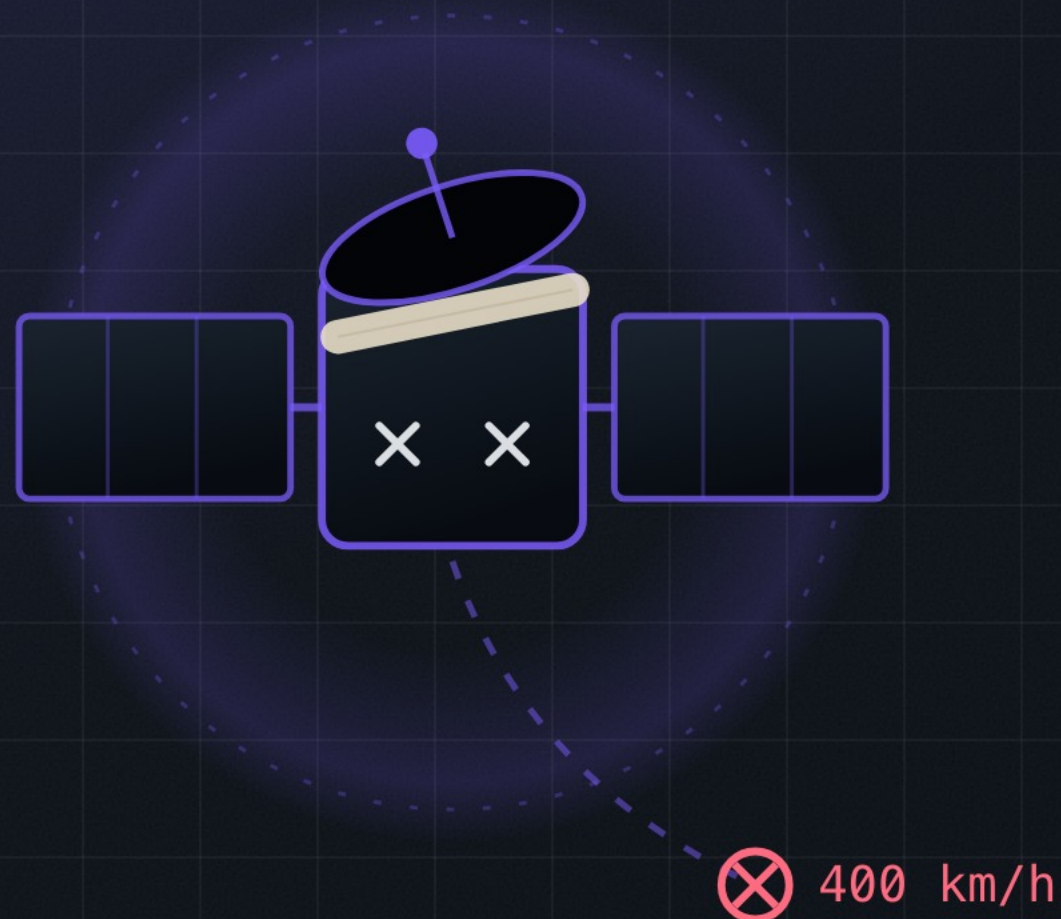


Swipe for why, and the fix.



// MEET THE WITNESS

Confident.
Cooperative.
Often wrong.



THE CONCUSSED WITNESS

exhibit 01

LocationManager

GPS looks like truth. It behaves like a witness who took a hit to the head, and it has never once suspected itself.



The Loopdown

@darkpandawarrior

02 / 06

swipe

Tunnels, towers, garages

In a tunnel it repeats the last spot for about 40 seconds, then jumps to catch up.

A big jump over a tiny time is your 400 kmph.



Three fixes

01 Catch the liars

if a reading needs rocket acceleration, drop it

02 Dead reckoning

estimate from the IMU and last good heading

03 Fuse, do not pick

weight each signal by how much you trust it now



// THE RESULT

50% to 95%

Trip accuracy, in production, on real drives.

Almost every point we were missing was a moment the phone lied with total confidence.



The Loopdown

@darkpandawarrior

05 / 06

swipe

Assume your inputs will lie

Good systems do not need perfect inputs.

Nobody gets perfect inputs.

They plan for the day the inputs lie.

